

No Matrik									

Name: _____

Group Tutorial: _____

KOLEJ MATRIKULASI PULAU PINANG
KEMENTERIAN PENDIDIKAN MALAYSIA
PENANG MATRICULATION COLLEGE
MINISTRY OF EDUCATION MALAYSIA

PRE-PSPM 1
PROGRAM MATRIKULASI SISTEM EMPAT SEMESTER
PRE -PSPM 1
MATRICULATION PROGRAMME FOUR SEMESTER SYSTEM

SAINS KOMPUTER 1
1 ½ JAM

JANGAN BUKA SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO.

ARAHAN KEPADA CALON:

Kertas ini mengandungi 6 soalan.
Jawab **semua** soalan di ruang yang disediakan.

INSTRUCTION TO CANDIDATE:

This booklet consists of 6 questions. Answer **all** questions in the space provided.

Untuk Kegunaan Pemeriksa		
No. Soalan	Pemeriksa	KKP/KP
1		
2		
3		
4		
5		
6		
JUMLAH		

Kertas soalan ini mengandungi 10 halaman bercetak.
This booklet consists 10 printed pages.

1. (a) Elements in Figure 1 represents a file, program, web page, or command. It is also useful to recognize quickly an object in a browser list.

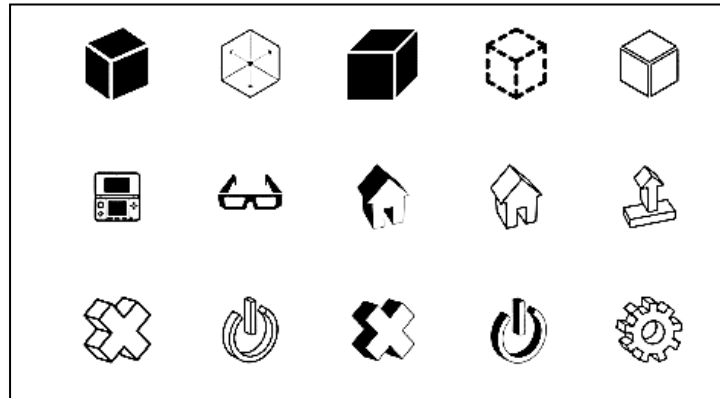


Figure 1

- (i) Describe multimedia elements in Figure 1.

[2 marks]

- (ii) Explain the purpose of multimedia elements in Figure 1.

[2 marks]

- (iii) Identify and describe the axis and dimension of the multimedia representation in Figure 1.

[2 marks]

2. (a) The process of designing and developing a multimedia product is essential to ensure a good quality of those products.

- (i) There are 3 phases involved in designing and developing multimedia. Briefly describe the second phase of the multimedia design.

[2 marks]

- (ii) Based on the description below, identify the Multimedia phases.

[4 marks]

Description	Multimedia Phases
Murugan has to make sure that the end product is at par with the storyboard.	
Mohsin needs to identify how the user will interact with the interface of the multimedia product.	
Marina needs to ensure that the multimedia product is suitable in any type of devices such as laptops, tablets and personal computers.	
Norli is incharge of converting non digital materials into digital materials	

- (iii) There are 3 types of authoring tools in multimedia design and development. Differentiate any two (2) of the authoring tools.

[4 marks]

Multimedia authoring tool 1	Multimedia authoring tool 2
Example:	Example:

(iv)

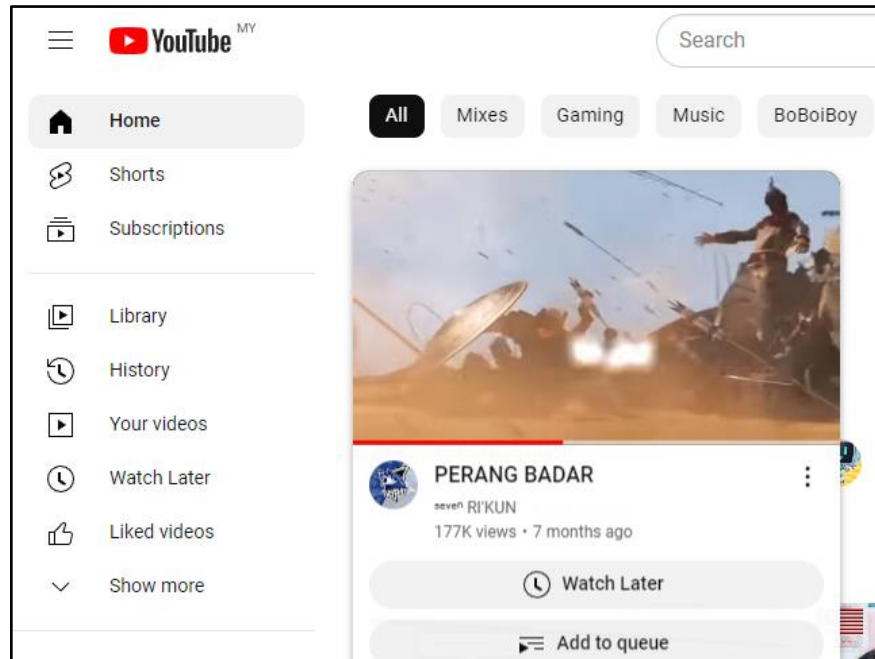


Figure 2

Identify the type of navigation that is shown in Figure 2.

[2 marks]

A	
B	

- (b) Establishing a photography business requires about two-thousand decisions right off the bat. Exactly how are you going to get those digitals from your computer to your client?

- (i) Differentiate the methods on how deliver digital images to clients

[4 marks]

Method 1: _____	How:
Method 2: _____	How:

3. File based systems were an early attempt to computerise the manual system before database system was introduced.

(a) Differentiate between file processing system and database system.

[2 marks]

File Processing System	Database System

(b) State one (1) ineffectiveness of file processing system that has caused the system to be replaced by database system.

[2 marks]

(c) Describe one (1) advantage of using a database system.

[2 marks]

4. A database object is any define object in database that is used to store or reference data.

(a) Explain the function of database objects below.

[2 marks]

Database Object	Fuction
Query	
Table	

(b) Identify one (1) more object other than 4(a).

[1 mark]

(c) State the function of database object in 4(b).

[1 mark]

5. (a) In information technology, a network is defined as the connection of at least two computer systems, either by a cable or a wireless connection.

(i) Give one (1) advantage of having a network

[2 mark]

- (b) In our daily lives, we have become so used to technology that it seems a natural part of our existence.

(i) State one (1) network use in health care.

[1 mark]

(ii) Describe your answer in 5(b)(i).

[2 marks]

- (c) Network media refers to the communication channel used to interconnect nodes on a computer network.

(i) State one (1) difference between wired and wireless networking media.

[2 marks]

Wired	Wireless

(ii) Give two (2) examples of each of the networking media in 5(c)(i).

[4 marks]

Wired:

Wireless:

- (d) Diagram 1 below shows a basic network diagram.

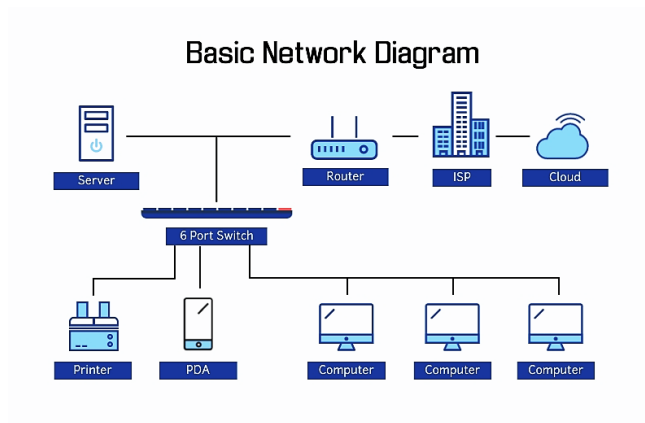


Diagram 1

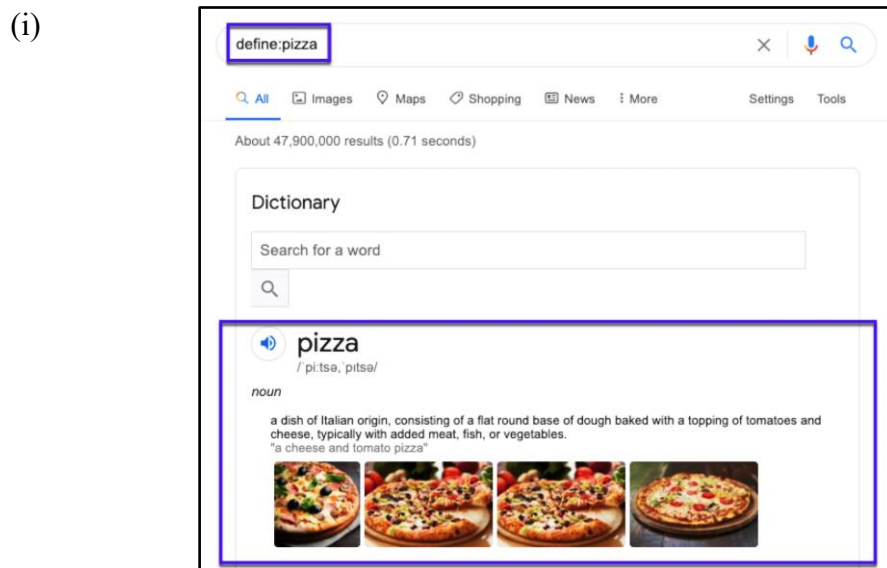
- (i) Describe the function of switch in the network above. [2 marks]
-
-
- (ii) Give one (1) reason why router is used in the network above. [2 marks]
-
-
- (iii) Identify one (1) more network device that is used to build a network besides switch and router. [1 mark]
-
- (iv) State the function of the device in 5(d)(i). [2 marks]
-
-

6. (a) Your supervisor at an Italian Pizzeria asked you to find out what the most popular pizzas are this year. He wants to have those ingredients stocked in the store for the upcoming holiday season. A) In order for you to find out that information, you turn on a computer and get ready to conduct a search on the internet.

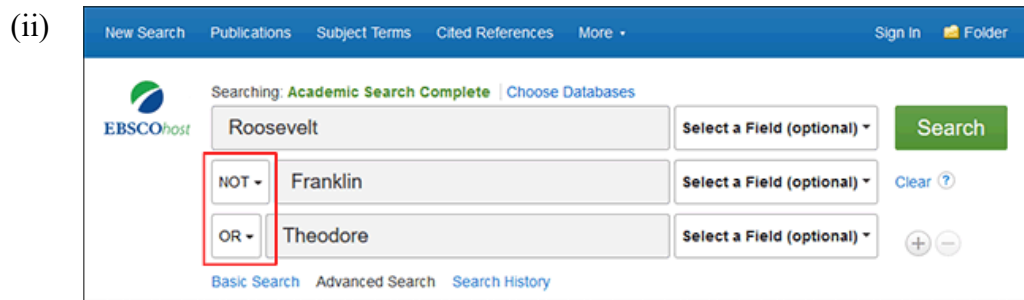
(i) Identify the most suitable type of search tool to be used in the scenario. [1 mark]

(ii) Describe the type of search tool mentioned above. [2 marks]

- (b) Search strategies will help you to narrow down the list of hits to more manageable numbers so that the searching process can be more efficient and effective. Identify the search strategy used in the picture given below.

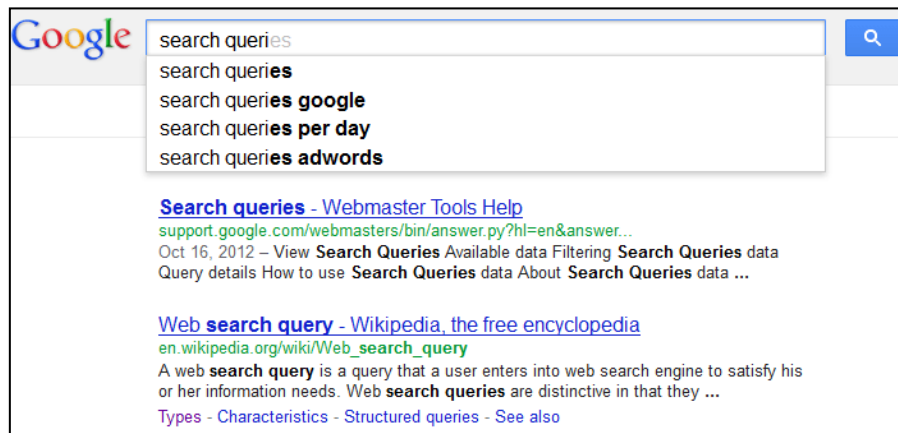


[1 mark]



[1 mark]

(iii)



[1 mark]

(c) Which phrase below will display less result?

“Computer Science” or Computer Science

Explain your answer.

[2 marks]

(d) Nearly all Internet users go online to conduct some of their ordinary day-to-day activities, from mundane tasks to social arrangements to personal recreation.

(i) Identify the types of web activities based on the descriptions given below.

[3 marks]

Description	Web Activity
Shopee, cimbclicks, lelong.my, OctaFX	
Google Classroom, Google Books, Google Scholar	
Astro Awani, Utusan, Malaysiakini, CNN	

Table 1

(ii) Describe one (1) web activity that you use daily other than mentioned in Table 1.

[2 marks]

- (e) Teenagers use digital technologies for everyday activities like keeping in touch with friends on social media, relaxing and doing schoolwork. Because they're online so much without parents' supervision, teenagers need to be able to identify acceptable and unacceptable online content independently.

- (i) Identify the issues that these teenagers may face.

[3 marks]

Scenario	Issue
Have their personal data collected using cookies	
They are exposed to contents such as pornography, violence, hate and terrorist sites.	
Get inappropriate marketing messages or scam emails	

- (ii) State two (2) ways to improve privacy when using the Internet.

[2 marks]

- (f) 4IR is changing how we live, work and communicate through the use of technologies like artificial intelligence (AI) and automation augmenting our professional lives, we're able to make smart choices, faster than ever before.

- (i) Explain the negative impact of 4IR in the job market.

[2 marks]

END OF QUESTIONS